

Section 3

Technology Building Blocks of Digital Transformation

Chapter 1 Introduction

Key Takeaways

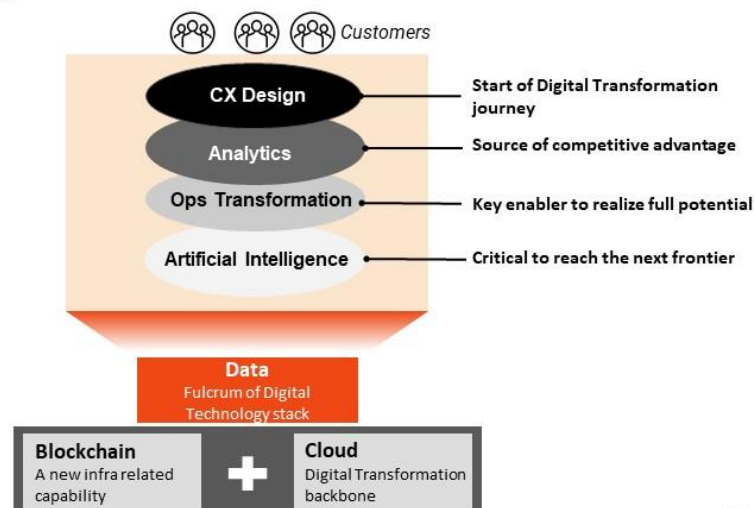
What must Experience Design incorporate to fuel successful digital transformation initiatives

Trends in Customer Experience

Lesson 1: Seven key building blocks of the digital technology stack

End-to-end technology orchestration required for successful Digital Transformation

Digital Fundas
A CSR initiative of Incedo



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The digital technology stack further has seven key building blocks. These are Design/Customer Experience, Analytics/Data Science, Automation/Operations Transformation, Artificial Intelligence (AI), Data Infrastructure, Blockchain, and Cloud. These play a critical role in the execution of digital transformation; they are the collective force behind successful digital programmes.

Digital transformation has to be end to end

The most important aspect of these technology building blocks is that they are not independent but are interconnected. Enterprises realize transformative value from them when they are connected end to end. These building blocks represent different layers of the technology stack; connecting them tightly creates a virtuous cycle.

Digital must start with design thinking to define strategy and customer experience. The whole experience must be thought up from scratch and redesigned to deliver 'what it should be' rather than starting from 'what it is'. One of the distinctive aspects of the digital customer experience is that it can be highly personalized and targeted at the individual. This is enabled by analytics. Analytics also enables customer insights, analysis of sales performance and operational efficiencies, and informs other key business decisions that help drive the business.

The superior customer experience that digital enables has to be fulfilled by internal operations. Typically, digital requires significantly improved response time, and this requires a redesign of processes, which is enabled by automation. Artificial intelligence (AI) can power enterprises to reach the next frontier of digital transformation. The self-learning ability that AI provides can sharply enhance both customer experience and operations efficiency over time.

The bedrock of everything is Data. All these technology building blocks require high-quality and deep data. Data is also where the recursive relationship or the virtuous cycle between the technology building blocks happens.

Digital is leading to a data explosion – in volume, variety, and velocity. However, to realize the full potential of this 'digital data deluge', enterprises have to upgrade and build scalable data infrastructure. This is easier said than done. Most enterprises struggle with getting high-quality, harmonized data together.

Cloud, which began as an infrastructure capability helping provide scalability and making costs variable, is now becoming a key enabler for transformation across the entire technology stack that we have mentioned. Cloud's scope and capabilities are increasing tremendously, and many enterprises are transitioning to a cloud-native architecture. It provides them a great opportunity to upgrade the various elements of the technology stack and, in particular, to address the data infrastructure and quality issues.

The various technologies that are the building blocks of digital are like the parts of an orchestra. Each of the parts needs to be very good, but only when they sync together that beautiful music happens. That beautiful music, in our case, is *realizing the full potential of digital transformation!*

Section 3

Technology Building Blocks of Digital Transformation

Chapter 2

Customer Experience Design

- Web/mobile interactions are necessary but not sufficient by themselves to deliver the gold standard in customer experience. The digital journey must progress beyond the first touchpoint of the consumer to the entire life cycle of experiences. Digital natives have done this successfully in various industries; legacy enterprises are expected to achieve the same.
- Customer experience is the new face/promise of brands that sets the winners apart from the losers. Driving higher customer loyalty is the key to generating shareholder value in the digital age.
- Successful transformation programmes take a holistic approach – defining the experience, one that puts the consumer at the centre, rejigging the organizational processes to deliver the experience seamlessly, and enabling end-to-end integration for consistently delivering high-quality experiences.

Key Takeaways

What must Experience Design incorporate to fuel successful digital transformation initiatives

Trends in Customer Experience

Lesson 1: Trends in Customer Experience Design

Experience Design - Starting point of the digital transformation journey



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Digital technology has transformed consumer habits and has caused a shift in customer expectations, resulting in a new kind of modern buyer. These buyers rate organizations on the basis of their customer experience and, increasingly, their digital customer experience first. And these digital experiences are no longer limited to interaction with the enterprise's website but have expanded in terms of channels and touchpoints to cover all aspects of the customer journey.

The benchmarks for these digital experiences are being set by the digital natives, including the FAANG companies. Customers are expecting similar or better experiences from all businesses they interact with. When they search for products or services, they want the search to be as intuitive as Google's; they want product recommendations as personalized as from Netflix; they want the checkout experience to be as effortless as on Amazon.

Given that experience design is the starting point, successful digital transformation initiatives need to include:

- Experiences designed to be intuitive, personalized, and with the customer at the centre.
- Organizational processes aligned to deliver world-class experience – seamless integration of the enterprise processes like onboarding, billing, supply chain, customer service, etc., using technology and AI/ML.
- End-to-end integration with partners, suppliers and other ecosystem players so that high-quality experience is delivered across all touchpoints.

Customer experience designs will be shaped by the following trends over the next three to five years:

Companies that have simplified customer experience are winning by a huge margin

- Customer attention spans have shrunk massively – they are looking for experiences that are simplified and transparent, and which do not hide policies and hidden charges behind fine print or have complex terms and conditions.
- According to a study by [simplicityindex.com](https://www.simplicityindex.com), the stock portfolio of companies that topped the customer experience simplification index (top 10) outperformed the major indices by 679 per cent between 2009 and 2018.¹
- Simplification requires deep understanding of customer needs and personalizing experiences to those needs. To achieve this, companies need to:
 - Create a unified view of the customer across different touchpoints and businesses
 - Convert the huge amount of data they collect into meaningful insights
 - Integrate the insights with systems of engagement on a real-time basis

Customers are looking for seamless omni-channel experience

- This holds true regardless of how your customers choose to interact with your brand, be it via the web, a mobile device, social media, IVR, a call centre, chat, or any other available channel.
- Customers want end-to-end experiences that are unified. Individually optimized point-of-contact solutions (e.g., sales, billing, service, etc.) that are disconnected and require customers to hop across multiple systems/applications will not work. Unified solutions will require integration across the enterprise.

Voice-first interactions are disrupting the customer experience

- Voice is a more effective way for humans to communicate instructions – voice commands are more efficient than navigation menus, allow for multi-tasking and, with the help of advances in natural language-processing AI/ML, can work with less-than-precise instructions, making it more accessible to customers (e.g., there will be no abandoned searches only because the user is not able to spell the words correctly).
- However, in many situations voice as an output mechanism is less efficient than screens. This is leading to convergence of pure voice-based systems (e.g., Amazon Echo or Google Home) and screen-based outputs (e.g., Google Assistant) to hybrid solutions like Echo. Echo uses voice first and then uses GUI-based interaction sparingly when needed.
- Further convergence will focus on the right balance between voice- and screen-based interactions as companies experiment with and learn to use the right mix for different use cases and customer segments.

Virtual and augmented reality is changing how customers engage with companies and employees perform their tasks

- VR and AR started more as a fad limited to gaming and video applications. However, with advances in the technology, companies are looking at reimagining processes and experience using VR & AR.
- Remote working, digital twins, paperless environments and hands-free working environments are just some of the applications that are transforming enterprises as both VR and AR hardware and software become more intelligent as well as more cost effective.
- With increasing applications in corporate training, customer onboarding, warehouse operations, logistics, etc., companies are increasingly turning to augmented reality solutions to overcome cost and distance disadvantages. We expect these solutions to get an even bigger boost as a result of the constraints on human-to-human interactions on account of the Covid-19 pandemic.

Experience design is extending from B2C to B2B and also to enterprise use cases

- The bulk of customer experience design thought has so far been focused on consumer-led interactions and businesses. However, increasingly companies are applying these design principles to B2B customers and also to internal enterprise processes and systems.
- There is a growing realization that customer experience is not just limited to website/app usage or interaction at a store POS (point of sale). Every function of an organization can impact customer experience, directly or indirectly. Designing effective enterprise processes that allow employees to perform their tasks in a timely and efficient manner can drive down costs and deliver better products and services to customers.

Intersection of data science, technology and UX design is key to designing world-class experiences

- The most effective experience designs emerge when left-brained and right-brained approaches come together to develop a well- rounded understanding of consumer behaviour, combining classical consumer research with advanced propensity models.
- Companies that test and refine their designs continuously perform better than companies that invest in massive design efforts upfront. Digital natives like Google, Facebook, Amazon, etc., invest heavily in their A/B testing infrastructure to run thousands of experience design tests simultaneously.

Nitin's conclusion:

Customer experience will continue to be the key difference between winners and losers in this Digital Age. Digital transformation is one of the most powerful tools an organization has at its disposal to differentiate itself on the customer experience front. This will require elevating the role of customer experience design within the organization to make it a first-class citizen. Organizations that do this effectively will reap rich rewards in terms of customer loyalty and shareholder value.

Section 3

Technology Building Blocks of Digital Transformation

Chapter 3

Used properly, Analytics is a Source of Competitive Advantage

- Digital experience and its promise of personalization can be delivered effectively only when analytics informs the 'man + machine' agents to engage with consumers proactively in the digital age.
- So far, analytics has been deployed by traditional enterprises to add incremental value, as opposed to the way digital natives have deployed it, which is to truly make it a source of competitive advantage.
- There is a wide gap between the promise and reality of analytics. This occurs due to six key reasons, which I outline in the relevant chapter. These must be carefully addressed by enterprises to make analytics a source of competitive advantage for themselves in the digital age.

Key Takeaways

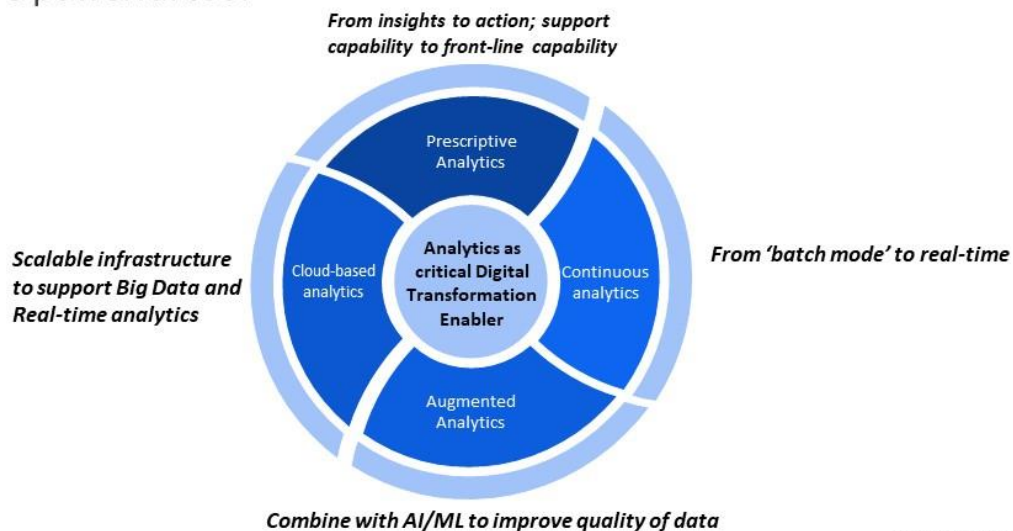
Trends in Analytics

Analytics: Gap between promise and reality

Future of analytics

Lesson 1: Analytics is changing dramatically, becoming an even more powerful tool

Analytics - Changing dramatically and becoming an even more powerful tool



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Analytics has grown tremendously in scope and scale over the past twenty years that I have been associated with it. However, I feel we are still seeing only the tip of the iceberg. With the continued explosion in data and rapid advancements in technology, multiple advancements are happening in analytics, which will take its impact to a totally different level, making it an even more effective enabler for enterprise digital transformation journeys.

I want to call out four trends, in particular, that promise to significantly raise the impact of analytics:

- **Prescriptive analytics.** Analytics has progressed from descriptive (what happened) to predictive (what will likely happen) to prescriptive (what should be done). This progresses analytics from insight to action, and from the realm of a support capability to a front-line capability. Machine learning is a key enabler in this progression of analytics from insight to action.
- **Continuous analytics.** One of the biggest trends in data has been the explosion of real-time data. This is moving analytics from 'batch mode', where analysis was done offline, to continuous analytics and insights. This trend is mainstream in both consumer and industrial situations. IOT is a significant driver of this trend; it can produce millions of real-time signals enabling dynamic tracking and optimization.
- **Augmented analytics.** This is about combining analytics with AI and machine learning, and automating tasks that might be traditionally done by a data scientist or a specialist analytics professional. One of the biggest leverages of augmented analytics is improvement of data quality. Often, more than 80 per cent of time and costs expended on analytics projects goes into ensuring adequate data quality. AI and ML can play an important role in improving data quality, which can significantly improve both the effectiveness and efficiency of the analytics process.
- **Cloud-based analytics.** Cloud is one of the biggest technology trends of our times and I believe it will have a profound impact on how analytics is done. Cloud platforms, of course, provide a scalable infrastructure for computing that can support and enable big data and real-time analytics. More than that, cloud platforms can help streamline the

process of gathering, synthesizing and processing data. This can be a major productivity booster. Additionally, cloud providers like AWS, GCP and Azure are making tremendous strides in the capabilities of their analytics platforms. Many of the tasks historically done by the analytics specialist are in-built in these platforms. This is a fascinating development and will have far-reaching implications for analytics specialists, in terms of the roles and skills required of them.

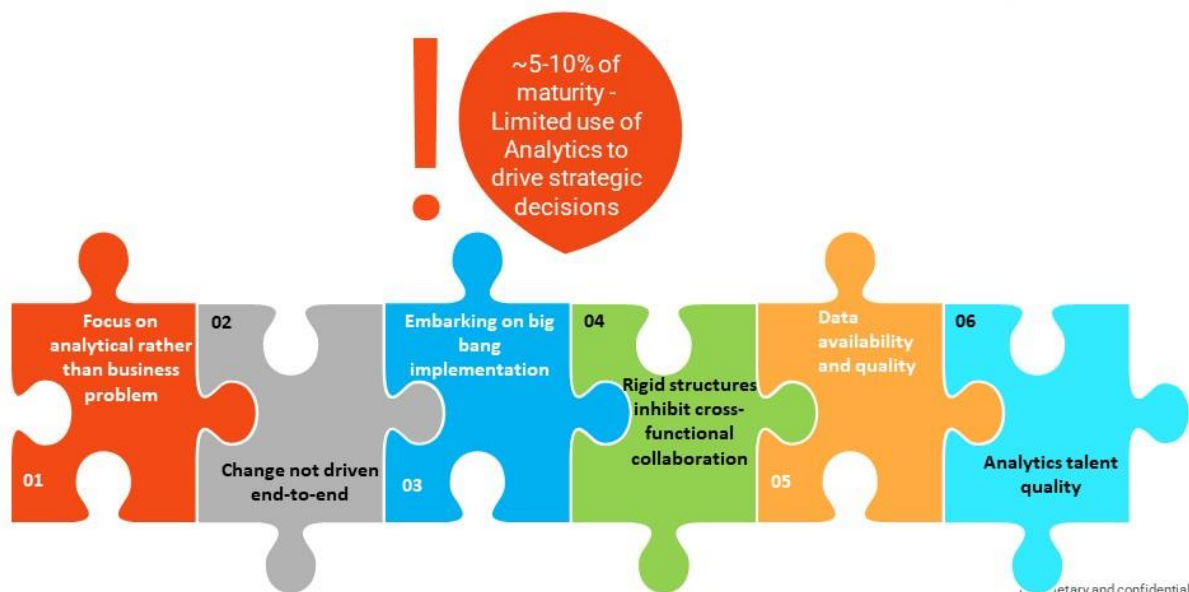
Nitin's insights:

I have been closely involved in analytics for the past twenty years and I am very happy to see it becoming a mainstream capability – an integral building block of any digital transformation. This was not always the case. I started my analytics journey at the McKinsey Knowledge Centre in 2002. When I moved from the consulting office to the knowledge centre, I was amazed to see how the combination of increasing data depth and analytical tools made possible analysis that was well beyond the scope of traditional consulting teams, even though the teams were staffed with talent of a very high calibre. I saw the possibility that analytics could become a core service line and potentially a source of competitive advantage for a consulting firm like McKinsey. Evangelizing the potential of analytics at that stage was not easy.

It was seen as a super-specialization, the domain of 'nerdy Phds'. It took many years of sustained effort and internal selling to establish analytics as a full-fledged service line within McKinsey. I am very happy to see that so many firms have embraced analytics as a core capability that it has moved from the sidelines to centre stage. It has caught the imagination of enterprises and the public at large, also becoming a source of significant job creation.

Lesson 2: Gap between promise and reality – 6 reasons

Analytics - Six reasons for gap in promise and reality



There are six main reasons why most analytics initiatives fail to deliver the value they promise:

1. **Focusing on analytical rather than business problems first:** In most analytics projects, there tends to be a gap between understanding of the business context and its translation into the analytics problem that is being solved. Analytics projects tend to overly focus on solving an analytical problem very well, when instead they should focus on solving the business problem by understanding the linkage between business and functional drivers. In my experience, data scientists are often attracted towards intellectually challenging problems and usage of the most sophisticated techniques. However, solving hard problems is much less relevant than solving problems that really matter. In the problem-solving process, problem discovery and scoping are a lot more relevant than analytics itself. Therefore, every analytics exercise must start with a sharply defined business problem based on specific use cases and business key performance indicators (KPIs). This can be achieved by the upfront involvement of business teams in problem definition and solution design.
2. **Change not driven as end-to-end transformation:** There are four different layers in analytics. The first is the strategy layer, which is about identifying the core business problems; the second is model building, which hinges on choosing the right kind of data modelling for your specific problem; the third is the actual data (access and collection), and the fourth is intelligent applications. Where many data analytics projects fall short is in not recognizing that to achieve true business impact, change needs to be driven as an end-to-end transformation, where business-problem definition, analytics modelling, the data platform, technology and operations implementation are tightly integrated. A disproportionate focus is often placed on the outcomes of modelling and not enough attention is paid to the core building blocks to connect all the four layers. I have seen many companies where analytics teams have hundreds of models but nobody is using them. Data scientists often derive great satisfaction from building great models, but a model by itself serves no purpose. It is only relevant when it is solving a clear business problem. What is more, even if the model is well connected to a business problem, you have to ensure that you can operationalize it and put it into production. Companies need to adopt an integrated approach towards building their solutions; only then can they derive significant advantage from their analytics investments.
3. **Embarking on big-bang implementation rather than starting small:** Analytics problem solving is inherently iterative. It needs a fair degree of experimentation to get it right. You need to start small and then build up in a progressive manner. Starting with too large a problem makes specific actions, iteration and insights discovery difficult. As I have said earlier, there is no 'first time right' in analytical problem-solving. Trying to do a big bang implementation with large data sets that have a cross-organizational impact is most certainly a recipe for disaster.
4. **Rigid structures inhibiting cross-functional collaboration:** Most organizations today are structured linearly, with negligible cross-functional collaboration. Successful analytics projects need cross-functional teams composed of data scientists, architects, engineers and teams from business, operations and even finance, who can work through agile implementation cycles until they find the solution to meet their business needs. A mindset and culture that are not agile, and structures that inhibit cross-functional collaboration, are often the hidden causes of failure in analytics-driven transformations. Moreover, placement of the analytics function within the organization itself can create challenges. I see different organizational models for placement of analytics – within a business unit, as a standalone shared service, placed with the technology function, etc. Each of these models has its issues. Analytics placed within the business has the benefit of proximity to the business context and to the problem to be solved, but typically this position does not allow for scale to build depth and quality. Standalone analytics units often tend to be 'orphans' struggling for organizational prominence. Analytics units placed with technology have the benefit of being close to data but tend to get very distant from business context. Clearly, there is no perfect structural solution for analytics. Therefore, cross-functional teams and a culture that emphasizes the importance of analytics are key to success in the use of analytics.
5. **Data availability and quality are poor or non-existent:** Data is the lifeblood of analytics. Without the right type and quality of data, analytics is not possible. I have

mentioned earlier that 80 per cent or more of time spent on analytics projects can go into getting the right quality of data together. In some enterprises, data can become an insurmountable challenge. Either data is just not there, or its quality is very poor, or it is very difficult to access using the technology systems. I have seen that many data scientists tend to get completely immobilized if there is no data. That is where it is very important for analytics professionals to develop not just the skills to work with data, but also problem-solving skills to be effective when data is not there or is limited!!

6. **Analytics talent quality – inability to solve problems:** The last decade has seen a real explosion of jobs in the category of data scientist. However, there is huge variability in the skill profile and quality of data scientists. Analytics is a skill set where the gap between effectiveness of the top decile and the median can be massive. While analytics professionals tend to be very focused on analytical tools and techniques, my view is that the biggest driver of success would be their problem-solving skills. That is the biggest differentiator between top analytics professionals and an average one. Analytical tools and techniques applied mechanically and without sharp problem-solving rigour rarely gets you great outcomes. The challenge is that problem-solving skills are not easy to build in a programmatic fashion. These skills are built through coaching and nurturing, and require the right role models and enterprise culture. Every legacy enterprise I know has ended up building analytics groups but is struggling to attract the right quality of analytics talent. This inability to attract talent is not because of their unwillingness to spend top dollars but because they struggle to build the culture that will attract the top talent. This talent issue has a huge impact on the output of the analytics function in legacy enterprises, where there is often a lot of analytics activity but not any corresponding high-quality output.

Nitin's conclusion:

To sum up the challenges with respect to analytics, its implementation is not a silver bullet. Analytics projects are messy, complicated and iterative. Unless they are approached with an end-to-end business transformation mindset, they will fail to deliver the intended results.

Ability to connect the dots within analytics is not easy to come by. There are very few organizations who are currently able to manage each piece of the equation in unison to deliver true business impact.

Lesson 3: What does the future of analytics look like?

Three key hypotheses

First, analytics will go mainstream and become all pervasive and Analytics would potentially cease to exist as a standalone capability or function. Analytics capabilities as we know it will split into two parts. First part will merge with business functions, and analytical skills will become a core skill expectation for all business professionals across marketing, finance, operations, consulting, or any other function. Second part of analytics will merge with the technology function. Picture Data, AI and Cloud as intersecting circles and what we know as analytics today will most likely get subsumed within these intersecting circles.

Second, the boundaries between services and platforms in Analytics will blur. Analytics was initially more about services, involving deployment of high quality talent with strong quantitative skills on data-intensive problems. Over the past few years, a host of analytics based products and platforms have emerged. With the growth of Cloud, these platforms have become more

pervasive. And, with AI, there is a growth in several point solutions. This growth in platforms and products will continue thereby blurring the lines between product and services, as a result many of these platforms and products will work along with services.

Third, expectations from data scientists and analytics professionals will change significantly. So far, the main skill expected from them was statistical modelling. With the growth of analytical platforms, various aspects of modelling will get automated and simplified, and this will no longer be considered a core skill. My hypothesis is that we will see the emergence of two varieties of analytics professionals: those with deep domain knowledge and problem-solving skills, and others who have strong technology skills and work at the intersection of big data, AI, and Cloud.

Nitin's conclusion

I continue to find analytics to be a fascinating space. I am convinced that the best days of analytics are ahead of us and the scope of its role and impact will grow way beyond what we can see today. However, it is important to note that in the long term it's structure and form will be very different. This will have lots of implications, both for enterprises and for talent looking to make a career in analytics. There is a lot to look forward to in this space!!

Section 3

Technology Building Blocks of Digital Transformation

Chapter 4

Operations Transformation

- Delivering the promised customer experience typically requires an end-to-end overhaul of the business processes.
- Even today, many traditional enterprises are stuck with a control mindset and have cumbersome processes. This often leads to bureaucracy, slow speed, and thus friction for the customer.
- Enterprises need to go beyond incremental approaches to operations and build confidence by embarking on more transformational and end-to-end change programmes. The changes should include data-driven measurement of operations, intelligent automation that goes beyond RPA, and leveraging of AI for self-learning of processes.

Key Takeaways

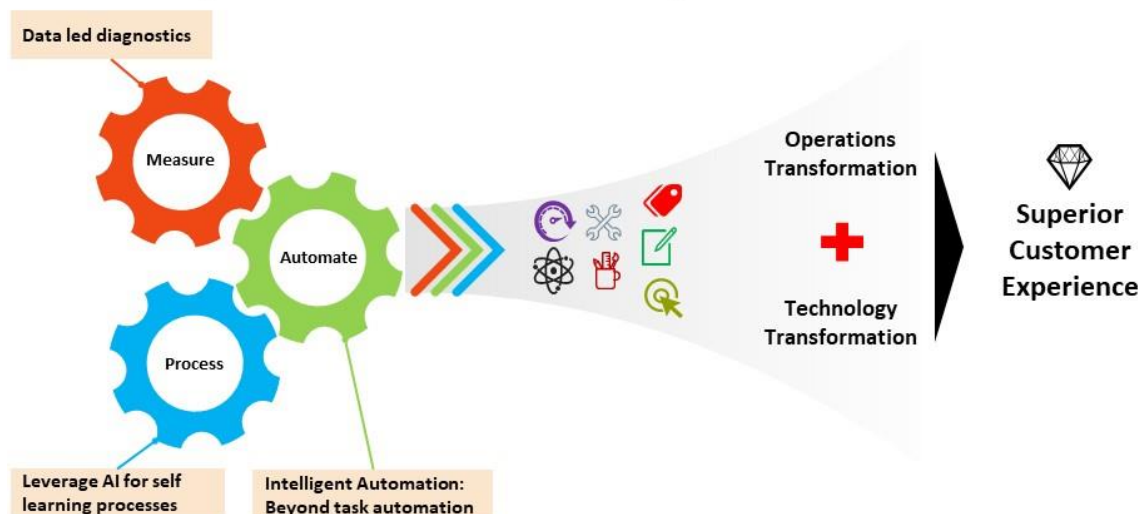
The next wave of Operations Transformation: Data, AI, and Automation

Lesson 1: The three trends that will drive the next wave of Operations Transformation

When operations transformation needs a complete overhaul

In every industry, the customer's digital expectations are rising, both directly for digital products and services, and indirectly for the speed, accuracy, productivity and convenience that digital makes possible. Organizations have realized that fulfilling these expectations takes a lot more than creating customer-facing digital properties: it requires a complete overhaul of end-to-end processes: from multi-channel fulfilment and customer support to billing and receivables. Amazon has upended the retail industry not by replicating the brick-and-mortar operating model but by reimagining the entire retail model with a digital-first mindset. While Amazon may have had the advantage of starting as a digital-native business, Walmart and Home Depot have successfully managed to integrate both their physical and digital channels to tremendous competitive advantage. One thing stands out for all the three firms: they approached the problem not in the narrow sense of digital customer experience but as a holistic end-to-end transformation, leading with operations as the main focus for driving their digital transformation. These firms realized early on that the traditional performance improvement initiatives – process re-engineering, cost reduction by piece-meal automation and outsourcing of operations – simply lack the required agility, flexibility and responsiveness to enable total transformation. In other words, operations transformation needed a complete overhaul.

Operations Transformation - Currently under-leveraged but necessary to realize full benefits of Digital Transformation



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- **Data explosion**, which will help drive process design and optimization using data-driven methods
- **AI advances**, which will help design and implement self-learning processes which can adapt to changing business needs
- **Automation maturity**, which will help organizations go beyond task automation and enable cognitive technologies

Nitin's insights:

Research clearly suggests that the future belongs to organizations who can make the transition to intelligent operations that leverage data and AI to develop and manage flexible processes and infuse agility and responsiveness in them. It is estimated that by 2030, banking operations will look very different: 75–80 per cent of transactional operations (e.g., payments processing) and upto 40 per cent of more strategic activities (e.g., FP&A, treasury) are expected to be automated.⁴ This is in addition to an even bigger structural transformation well underway: branch networks are shrinking at an ever faster rate; digital-only banks are jumping into the fray. The twin impact of these trends will mean that bank operations will be profoundly different from what they are today. And this makes it necessary for banks to address the operations transformation challenge on priority.

Data Explosion: Diagnostics and optimization reimaged

Edward Deming, the doyen of TQM, was reported to have said, 'You cannot improve what you cannot measure.' In the pre-digital era, process monitoring involved laborious time and motion studies. Today, most organizations have systems (e.g., ERP, WMS and other transaction processing applications) that measure the flow of information across the enterprise. This has enabled the creation of 'digital twins' – of existing processes and their modelling for analysis and simulation – through process mining.

Process mining has emerged as a key technology lever in business process transformation. Data mining algorithms are applied to event log data recorded by the transaction systems. This will identify trends and patterns related to process-time delays, throughput rates and shifting bottlenecks. There are distinct advantages in doing this:

1. **Discovery of issues in execution:**
Identification of shifting bottlenecks from activity-level event data logs
Highlighting of quality issues and reworking in complex cross-functional processes
2. **Improvement in process compliance and auditing:**
Process mining acts as a quality control mechanism and provides data-backed near real-time evidence on process compliance for regulatory/auditory purposes
3. **Assessing impact:**
It helps assess the dollar impact of an improvement action
Provides granular visibility into process performance at both worker and session level
4. **Identification of automation opportunities:**
Process mining uncovers specific opportunities for RPA and ML-based process automation
Helps prioritize automation opportunities

AI: self-learning processes for exception handling

Google's AlphaGo defeated the world champion in the game Go, arguably the most complex game in the world. This was more than a research project: it has opened up the possibilities of leveraging deep learning methods to build self-learning engines for designing and operating processes. This could have remarkable implications for business operations, where a key source of inefficiency is exception handling. In other words, the scope of 'straight-through processing' can be expanded to cover a wider array of scenarios that don't need to be defined upfront, but the process can adapt and learn as it keeps evolving with real-life scenarios.

We are already seeing artificial neural networks (ANNs) being applied to solve process optimization problems. Inspired by the structure of the human brain, neural networks have the ability to train themselves to learn from data. In other words, you no longer need to codify the entire set of operating rules upfront, but let the AI engine evolve the rules as it keeps learning from real-life scenarios. There are several potential applications across various operational processes that would otherwise require human intervention, and where decisions are often made on intuition. Some examples:

- Logistics: risk management strategies across distribution channels, design and management of transport capacity,
- Human resources: management of performance and human behaviour strategies at the individual level, sizing of personnel structure allocation at economic organization level,
- Accounting: identification of financial frauds, audit procedures.

On the cusp of the third wave with intelligent automation, going beyond task automation

Automation has seen tremendous advancement in the last ten years. The first wave saw traditional automation that automated specific tasks (e.g., screen scraping to extract data, screen recorders, et al). These served to deliver incremental task-level efficiencies. We are currently in the second wave, where software bots, process design tools and workflow rule engines have helped automate specific tasks (e.g., auto-routing a customer care call to the right agent; rule-based PO approval processes). These continue to drive efficiencies in processes that lend themselves to well-defined flows based on predefined rules.

I believe we are on the cusp of the third wave of 'intelligent automation', with the industrialization of NLP, voice-to-text and computer vision, which rely on sophisticated AI techniques to draw valuable information from hitherto unstructured data sources. A comprehensive intelligent automation strategy needs to be executed at a functional process level with a combination of complementary technologies:

1. **Task Automation:** With tremendous advances in AI applications for processing unstructured data, there is opportunity to expand the scope of automation to include activities that were previously reliant on human agents:
 - a. **Customer Care:** Call transcription and classification can be done by a voice-to-text bot in real-time, leaving the agent to focus on customer interactions.
 - b. **Account onboarding:** Document scanning and extraction of customer information, signature verification, etc., can be offloaded to computer vision bots, freeing up operations bandwidth to focus on customer engagement.
2. **Intelligent BPM:** Business workflows no longer need to be straight-jacketed process steps. AI expert systems offer a way to define a skeletal process workflow, and as the system learns from real-life experience it continues to refine and evolve the workflow rules:
 - a. **Audit process:** AI bots can evaluate journal entry submissions in real-time and based on contextual information (e.g., who, when, where and how the entry is prepared and submitted) and flag them for appropriate review. The data associated with the entry may include emails and supporting documents in addition to the standard accounting rules. This has the potential to reimagine the audit process from a post-facto analysis to an in-process, real-time activity.
 - b. **Dynamic inventory management:** The traditional method of safety stocks at each level in the supply chain has always been hugely inefficient. AI-enabled systems are already making a big impact here by responding to signals across the supply chain in real-time and dynamically reallocating inventories to optimally manage service levels. These systems have the ability to learn and improve – and can further be tasked to optimize around specific goals (e.g., new product launch).
3. **Smart analytics:** As business complexity continues to grow, managers are expected to take decisions ever faster, often with limited information. Decision support engines – in the form of Descriptive, Diagnostic and Predictive insights and recommendations – are proving to be invaluable in helping humans make better decisions. Increasingly, we see decision-makers augmenting their intuition and experience with machine-enabled recommendations:
 - a. **Resource planning and allocation:** From capital resources (e.g., machinery, vehicle fleets etc.) to human resources, analytics can help drive their optimal allocation across priority areas. This, combined with a performance tracking and monitoring system, can help identify potential hotspots and enable focus on specific areas requiring intervention.
 - b. **Operational metrics:** Business operations teams are increasingly reliant on data-driven methods to define key operational parameters – from budgeting to daily operations monitoring in almost all functional areas. For instance, the customer onboarding team in a bank tracks 'speed to revenue' and, subsequently, CSAT as a key metric to plan and execute the onboarding process.

Nitin's conclusion:

It is clear that after the first wave of digital transformation of creating online and omni-channel customer experiences, the next wave of value is being created in operations transformation. And, as firms like Amazon, Walmart and Home Depot have shown, the truly successful organizations are the ones that are successful in integrating their front-end experiences with their back-end operations. To be sure, technology alone cannot be the answer – operations transformation will require significant focus on the overhauling of processes. Often, automating an inefficient

process would be worse than leaving it alone. So it is just as important to invest in people; operations, more than any other part of the organization, needs a blend of human intuition and the power and scalability of machines, calling for changes from training personnel to rewriting job descriptions, and so on. In other words, operations transformation is not easy, but then again, it is apparent that not doing it is no longer a choice either.

Section 3

Technology Building Blocks of Digital Transformation

Chapter 5

AI powers the forward march of Digital Transformation

- AI is becoming mainstream, especially among digital natives. However, it has not yet reached scale in many traditional enterprises as it is embraced mostly as a 'feature' and not as an 'architecture'.
- AI has the potential to drive the next frontier of digital transformation, both on customer experience and operations transformation. To make this happen, there are four key considerations, which I have outlined in the chapter on AI.

Key Takeaways

How can AI power digital transformation to the next level?

AI-enabled Digital Transformation: Execution imperatives

Lesson 1: How can AI power digital transformation to the next level?

Over the last few years, AI capabilities have really come of age; AI has matured from an experimental topic to a mainstream technology delivering significant value. As confidence in AI-enabled digital processes grows, enterprises need to push the boundaries of AI application to bring it to the complex core of their business processes.

Hyper-personalization across all interactions

In today's environment of digital hyperactivity, customers are spoilt for choice in products, services and content. While this sounds like true nirvana, there is a major flipside to it too. Given too many choices and decisions to be made, customers often make poor choices and end up being less satisfied than they perhaps were with fewer choices. And sometimes, being overwhelmed, they switch off completely when they have to make a choice.

Artificial intelligence-driven hyper-personalization, i.e., automatic curation of best choices for the customer, delivered in real-time across various channels of interaction, is the key to take digital customer engagement to the next level. For instance, as telecom customers look to take up new plans or upgrade their existing plan, they are presented with a seemingly infinite number of choices as to devices, plans, data plans, talk time, bundles, accessories, and so forth. It can be mind-boggling to really make a decision across these choices. However, telecom service

providers can leverage AI to analyse and uncover customer preferences, usage patterns and other key attributes to simplify the choices to the top three. This not only makes it easy for the customer but fetches her customer delight too.

Automation really becomes intelligent

While the focus of the first wave of digital transformation has been on customer engagement, the real value of digitally transforming the enterprise's underlying business processes – such as customer service, supply chain, business operations – is yet to be realized. Significant progress has been made in leveraging technologies like robotic process automation to bring some degree of automation to mundane, standard and monolithic processes. However, these technologies have been largely insufficient, given the complex processes that require decision-making intelligence.

Use of artificial intelligence to augment, automate and speed up decision-making in complex process journeys is a critical catalyst to really make automation intelligent and take it beyond the realm of simple process steps to automated complex decision making.

A case in point may be the back-office operations at any bank or wealth management organization, which are among the most critical and most resource-intensive processes. The complexity of managing different types of documents and information makes it extremely difficult for any RPA technology to automate it. AI and natural language processing capabilities are really pushing the boundaries of automation with intelligent text data extraction and interpretation, enabling analysts to make faster, better decisions, hence reducing the AHT (average handling time) for each request. In several cases, AI-driven capabilities can achieve full decision-making ability, making straight-through processing a reality.

Opening new AI frontiers – AI not just as digital technology but also for digital technology

Digital transformation is really converting every business into a technology business. With new technology-centred products, services and business models growing, so is investment in digital technology to support them across all layers – infrastructure, data, applications and security. Technology has become the core of investment, operations focus and business value realization. The opportunity costs of disruption or downtime of IT infrastructure, erratic behaviour of applications, poor data quality or even data theft are way too high to be taken lightly.

The potential to leverage AI to optimize the technology stack to best utilize precious resources, avoiding any downtime or disruption, and enhance the value realization, is very significant and real. The AI-driven approach is becoming key to efficient management of tech systems; e.g., automated monitoring and predictive maintenance of the digital infrastructure to ensure there is no disruption in the revenue-generating applications is as much a critical process as any other. AI algorithms can monitor the server logs, events, data traffic and other system parameters in real-time to automatically monitor the digital infrastructure – servers, devices, networks and applications – deliver early warning for proactive maintenance, and in some cases even auto-heal systems.

Similarly, while there is so much excitement around digital data, to really unlock its value it needs to be of high quality. Leveraging AI approaches to uncover data quality issues such as hidden data anomalies and help rectify them, will really take the realizable value from digital data to the next level.

As data and digital technology have become the lifeblood of the enterprise, the need for strong cybersecurity capability has never been felt to be more critical. With the growing sophistication of cyber threats, AI-driven approaches to automatically decipher and ward off unauthorized access attempts and network and system vulnerabilities, and recommend or auto-fix security issues can be extremely important to mitigate the risk of malicious attacks and data theft.

AI on the team – Human plus Artificial intelligence = Augmented Intelligence

As the digital transformation journey progresses, enterprises need to think about AI as the core member of the team. The debate of human intelligence vs. artificial intelligence is irrelevant. In order to get deeper below the surface of digital transformation, enterprises need to harness the complementarity of human and artificial intelligence.

As we rethink digital-driven operating models and business processes, artificial intelligence cannot be an afterthought anymore; rather, it must be at the core of it. The key is to think about how AI capabilities can be a performance enhancer for the employees, enabling better, faster decision-making and analysis of hidden data patterns which are humanly impossible to uncover.

Lesson 2: AI-enabled Digital Transformation: Execution imperatives

- **Leveraging artificial intelligence across the business stack** Enterprises must think about leveraging AI capabilities across the business stack – customer experience, business operations as well as the technology ecosystem. As they reimagine digitally-driven business processes, enterprises need to systematically curate opportunities across the business stack where AI can make a fundamental impact. The key lies in redesigning AI-enabled digital process maps, enabling better, real-time and autonomous decision making, driven by insightful recommendations and automation. Prioritizing such opportunities and implementing them iteratively will propel enterprises to achieve the next level of digital transformation success.
- **Embracing ‘AI as architecture’ and not ‘AI as a feature’** Artificial intelligence capabilities are no more to be thought of as augmentation to the existing technology stack, but rather, every component of the digital technology stack needs to be designed with the fundamental principle of AI-enabled real-time decision making. This requires rethinking the technology and data strategy. The underlying technology and data architecture need to get aligned to the requirements of AI-enabled processes. For instance, AI algorithms need access to a wide array of diverse digital data points (multiple formats, sources, high speed, large volume), to uncover hidden patterns and to make the right decisions. The underlying data technology architecture needs to have the capability to handle such a complex data ecosystem and process it real-time to power AI algorithms. Similarly, the underlying compute capacity of the digital tech infrastructure should be able to respond in real-time to the demands of processing this data for the AI algorithms to work.
- **Investing in and experimenting with new AI technologies** Artificial intelligence technologies themselves are evolving very rapidly. New AI techniques like convolutional neural networks, self-learning algorithms like reinforcement learning, cognitive intelligence technologies such as computer vision, are all growing in capability and unleashing new possibilities of application across business processes. Enterprises need to systematically experiment with these new technologies, test them in their business processes to assess their potential impact, and scale the successful ones as mainstream AI capabilities.

The culture of experimentation-driven innovation will help enterprises be at the forefront of leveraging AI to deliver consistent impact from digital transformation investments.

- **Rethinking the talent model in the AI-enabled world**
As I have said earlier in this chapter, enterprises need to harness the complementarity of

human and artificial intelligence. The manifestation of this complementarity is in the form of a talent model which will be relevant in the AI-enabled world. It requires skilling and training of employees to prepare and enable them to co-exist and grow with machines. The ability to work with or use AI is not just limited to highly skilled technology groups, it needs to become pervasive across all skill groups and levels in the enterprise. A customer service representative who can leverage AI-driven intelligence systems and its recommendations to resolve customer issues rapidly and to the customer's delight can make a far more significant impact in terms of business success. As the wave of digital transformation progresses, the ability of a company's employees to enhance their impact, by not just designing AI systems but also by using them, will be one of the most critical assets for a successful digital business.

Section 3

Technology Building Blocks of Digital Transformation

Chapter 6

The Digital Data Deluge - Boon or Bane?

- Data is the centrepiece of digital transformation, enabling design, analytics, operations transformation and AI. Without data, digital strategy cannot be executed.
- The exponential growth of data in volume and variety, and at unprecedented velocity, is the engine of success for the digital natives. At the same time, it is paralysing many traditional enterprises that have invested millions of dollars in it but have not realized the desired impact from it.
- It is imperative that enterprises develop a digital data architecture that is fit-for-purpose and also change their operating processes to harness the power of data in the digital age.

Key Takeaways

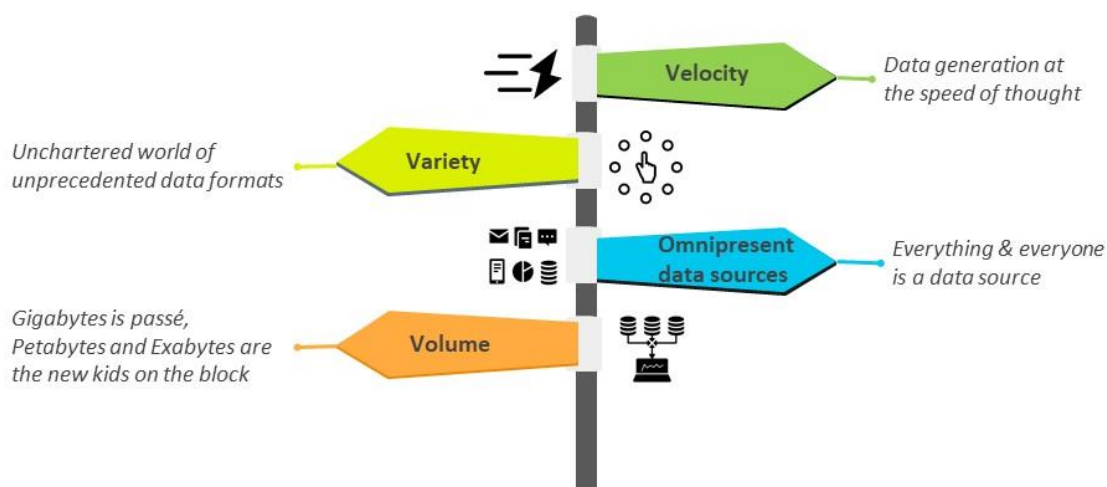
Changing profile of data in the digital age

Pressure points of the data ecosystem

Agility: The new mantra for enterprise data management

Lesson 1 : The changing profile of data in the digital age

Digital Data Deluge - Changing profile of data in Digital age



Introduction

Just when enterprises were starting to believe that they can handle their enterprise data and have the required management and governance ecosystem in place, they were hit by an unprecedented wave of technology-driven digital transformation. The new technology wave in the form of digitization has taken over the world, rendering carefully crafted data management frameworks obsolete. Suddenly, enterprises that were believed to have achieved data management maturity dropped down to the level of novices in their capability to handle digital data.

The digital wave, fuelled by massive volumes of complex, high-speed data – unstructured and uncurated – generated by mobile apps, social media, cloud and devices, has significantly impacted the enterprise's ability to manage and generate value from its data. The rise of the internet of things (IoT), new technologies like artificial intelligence, genome sequencing and augmented reality have made the challenges even greater.

The implications? Enterprises face a new, dynamic and complex digital agenda that is disrupting their data management and curation capabilities. This disruption is making it critical for them to consider the opportunities presented by digital transformation so that they may manage the unprecedented and new risks of digital data, which is arriving in waves that their walls cannot contain and which they have no control over.

The opportunities and risk? Digital data is a potential gold mine, and digital natives like Google, Facebook and Amazon have already demonstrated the impact digital data intertwined with AI and machine learning technologies can have on creating digital businesses and value where none existed earlier. Several leading enterprises have even started applying these principles and artificial intelligence technologies to their B2B data, which too is expanding rapidly. On the flip side of the coin are the potential risks – privacy, security and much more – that digital data produces for enterprises if not handled well.

The changing profile of data

As enterprises navigate the digital age, leveraging digital technologies like AI, machine learning, automation and cloud to create new digital opportunities, they need to realize that **digital data is the fulcrum of the digital technology stack and the centerpiece of digital transformation**. Enterprises need to reimagine their data management approach – aligning people, processes and technology – creating an agile, scalable, secure and, above all, digital-ready enterprise data architecture.

Much has been said and written about the trends of the digital age that are fundamentally changing the way businesses operate and the way consumers interact with them. These interactions and operating models are generating a very wide array of unprecedented data forms.

Examples include, but are not limited to:

- Connected medical devices, tele-health and consumer-driven healthcare.
- Genomic sequencing data and its use in precision and personalized medicine to make decisions about specific treatment paths that may be effective for the individual at hand.
- The internet of things, including machine-to-machine (M2M) integration and sensor technology in manufacturing, supply chain, telecom, oil and gas, and utilities, among other sectors.
- Omni-channel consumer choices in retail, consumer goods, banking and insurance transactions.
- Digital payments and transactions, among both enterprises and consumers, which are rapidly increasing across the world.

- Gamification – the introduction of game-like metrics such as rules, competition, prizes or points-earning – by an increasing number of banking, financial services and insurance companies.
- Enhanced, customized consumerism, such as use of facial recognition and biometrics, to create personalized experiences based on prior shopping behaviour and increasingly sophisticated predictive analytics.
- Digital marketing targeted at specific consumers based on their unique digital profiles and online behaviour.

The characteristics of these data forms are widely different from those of typical enterprise data like ERP, CRM, sales and finance, which enterprises and executives are familiar with. As we look closely at these digital data profiles, a few key themes emerge. It is very important to understand these digital data profiles and themes to be able to define and build data management frameworks for the digital age.

- **Variety – an uncharted world of unprecedented data formats** The rapid evolution of digital technologies is leading to exponential growth in new and unconventional data formats, such as radio frequency identification (RFID) tags, smart metres, global positioning systems (GPS) and machine-2-machine. This is all in addition to the clickstream data produced by our interactions with websites, mobile apps and other platforms. In the banking industry, as banks become more and more paperless, identity resolution will be achieved with retinal scans or image recognition rather than with documentation; physical signatures will be replaced by digital signatures; and paper cheques replaced by digitally signed e-cheques. The data infrastructure must learn how to handle, store and process these digital data forms (such as images, fingerprints, retinal scans, digital signatures and social media data), which go beyond the conventional customer demographic or financial transaction data.
- **Volume – Gigabytes is passé, Petabytes and Exabytes are the new kids on the block**

A single human genome takes up to 100 gigabytes of storage space, and as more and more human genomes are sequenced, data storage and processing needs will go from gigabytes to petabytes to exabytes. By 2025, approximately 40 exabytes of storage capacity will be required for human genome data⁵. While storage is one challenge, the bigger challenge is the ability of the current data infrastructure to process this data for 'signal' detection. To come closer to the consumer world, IDC predicts the average person will have nearly 5,000 digital interactions per day by 2025, up from the average of 700 to 800 currently. Such high-volume interactions will put extreme pressure on already struggling enterprise data ecosystems to process and curate the consumer data.⁶

- **Velocity – data generation at the speed of thought** On a daily basis, Facebook generates 4 petabytes of data, we send 294 billion emails, 65 billion messages over WhatsApp and make 3.5 billion searches on Google. That is the mind-boggling speed of event and streaming data generated across the board – from internet, consumer, financial transactions and IoT. 'Real-time' or 'high-speed' data processing and integration is required by enterprises to interact with and service their customers. The enterprise data architecture had just achieved some degree of efficiency in processing slow-paced batch data, and now suddenly they must adapt to need-for-speed.
- **Omnipresent data sources – everything & everyone is a data source** In the digital ecosystem, almost every device and every human being is generating huge amounts of data. Every interaction is generating a wide array of rich data covering multitudes of behaviours, interactions, conversations. Every person generates an average of 2.4 Gb of digital data each day, in the form of emails, messages, transactions, browsing, searches, videos, photos, health, vitals, biometrics and much more. Every device – industrial, sensors, wearables, computing – is producing data of varying forms – temperature, pressure, geolocation, biometric, logs, events, and so on. The sources of enterprise and external digital data have become unmanageably numerous and are ever increasing.

Lesson 2: 'Pressure Points' of the data ecosystem

The current enterprise data ecosystem has many pressure points.

- **Inability to consume, process and integrate high-speed, complex digital data**

The monolithic data architecture of traditional data systems lacks the ability to ingest and store data from different sources. They don't have sufficient processing capabilities to process various data forms – relational, non-relational, text, streaming, etc. Many organizations have taken steps to augment or migrate to big data technologies to overcome this challenge, setting up on-premise data ecosystems. It worked well for a while, but when the volumes started growing beyond the capacity they could handle, even the big data ecosystem didn't prove sufficient to fulfil the needs of the demanding digital data.

- **Limitation in curating and differentiating 'signal' from 'noise' – enabling AI/ML-driven insights**

The greatest value in digital data lies in its ability to enable deep insights, using artificial intelligence and machine learning, as to behaviours, patterns, recommendations, early warnings, next-best actions and much more. This allows for highly personalized, real-time actions and informed business decision support. However, the rigid information architecture of enterprise data systems focuses on standardization of data models and analytical tools rather than enabling dynamic business analytics demand. It is, therefore, ill-equipped to generate AI/ML-driven insights from digital data. These systems don't have the ability to curate such complex, large volume data in real-time to really differentiate and extract valuable signals for business actions and personalized customer interventions.

- **The 'golden record' is no more sufficient – digital interactions require 'golden network' for holistic 360-degree view**

Enterprises have lots of data, but it is siloed and disconnected. Data is spread across departments (CRM, sales, marketing, product development, customer service), product lines, platforms (web team, data warehouses, NoSQL, data lakes), locations (cloud, on-prem, edge, IoT, mobile) and systems. This complex ecosystem creates disconnected data, which leads to problems. A holistic or 360-degree view of its users, customers, products, accounts, partners, location and more is at the heart of leveraging value from digital for the enterprise. Most MDM systems, built on an RDBMS backbone, focus on creating the 'golden record', and are proving inefficient in creating a unified 360-degree view. Even with advanced computing and high-speed networks, the performance and ability of such systems continue to lag requirements. Why? Because it is no longer sufficient to build a golden record. As digital data is now an interconnected network of entities and relationships, the modern MDM system needs the ability to connect and correlate different data domains uncovering hidden relationships and insights.

So, while current MDM approaches continue to solve monolithic master data use cases, they have proven ineffective in serving connected data use cases.

- **Growing sophistication of cyber-attacks and data theft have redefined data security demands**

Digital transformation is all about turning data into knowledge, and the stakes are ever higher as it brings about a data-driven and connected world. The flipside of this transformation is the increasing risk of data breaches and cyber-attacks, which can cripple even the most resilient of organizations. Additionally, new regulations, on the level of personal data protection, such as the General Data Protection Regulation, increase the costs of data breaches even more.

With distribution of data across locations (cloud, on-prem, edge), managing the security and privacy of critical customer and enterprise data has become very difficult. The security frameworks which worked very well in securing data within enterprise walls will need to evolve to be secure with digital data while the enterprise also takes advantage of the business opportunity presented by it.

- **Poor data governance and quality are proving to be among the most critical challenges to truly realizing value from digital data and artificial intelligence capabilities**

Lack of data quality and enterprise data governance are proving to be the top hurdles in harnessing value from the digital data deluge. Easy and transparent access to high-quality enterprise data assets is critical to enable digital and AI/ ML-use cases which are worth millions (and many a times billions) of dollars. Data governance is one of the top differentiators between firms that have achieved the expected value from data and firms that have not.

While several organizations have attempted to achieve data governance maturity, most are still struggling, for a multitude of reasons ranging from incomplete understanding of digital data domains, inefficient or missing enterprise data architecture, poor metadata management and lack of robust data quality management processes.

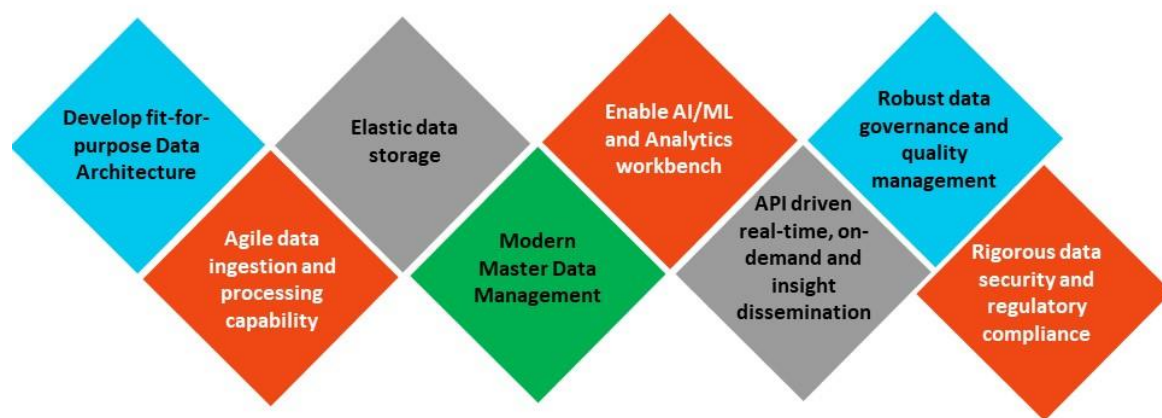
- **Real-time and on-demand availability of insights whenever and wherever required**

The enterprise data ecosystem should be able to provide 'real-time' or 'on-demand' access to insights and recommendations to make the best possible offers to the customer while she is shopping, deliver real-time alerts regarding a potential fraudulent financial transaction or warn a machine operator about potential maintenance issues in his machine.

The insights dissemination mechanism of the existing data ecosystems is not designed to deliver such insights instantaneously or at the most appropriate times. They are designed to deliver reports and insights to business executives at a fixed frequency and at a fixed time, and not necessarily to auto deliver them at the 'right time' and 'right place' where action is being taken.

Lesson 3: What are the key agile design principles enterprises should consider for digital data architecture?

Data - Key design principles for leveraging the most from Digital Data



Proprietary and confidential

Introduction: Digital data architecture – agility, and not maturity, is the new mantra for enterprise data management in the digital age

Most enterprises yearn to achieve data management maturity and keep making investments towards this end objective. The fundamental flaw in that approach is the underlying assumption that the target state (high maturity level) is static. The reality of the VUCA world of digital transformation, and hence the digital data generated by it, is diametrically opposite to that assumption.

In the uncertain and complex world of digital data, *agility (and not maturity) is a more effective fundamental design principle* to achieve the promised potential of digital transformation. As the digital wave progresses, there are several more complex data scenarios to be uncovered, even while those we already know of are being handled. Hence, the digital data architecture of enterprises has to have the capability to handle all the varied scenarios of today, but also the agility to quickly adapt to the ever-dynamic digital data landscape.

Key Agile Design Principles

1. **Data stacks don't need to be monolithic; they need to be aligned to data use cases**
Different stacks (traditional RDBMS, open-source big data or cloud-native) can co-exist and complement each other. The key is to identify the use cases for which different

stacks are best suited. Managing large enterprise data coming in from ERPs, CRMs and other operational systems, which is largely structured, relational and predictable, is best done by on-prem data systems leveraging well-tested data warehousing technologies. However, managing dynamic and complex digital data and leveraging artificial intelligence to derive insights from it requires modern stacks which are flexible, elastic and high-speed. Hence, the answer to the complex question of digital data architecture lies in the complementarity of different data technology stacks in a hybrid, fit-for-purpose ecosystem.

2. **Agile data ingestion and processing capability with rich interface using external data ecosystem**

Event-streaming data is what really epitomizes digital data profiles – it is fast, complex, high-volume. Event-streaming data is far greater and faster than the standard, transaction (ERP- generated) data. In addition, the data formats are diverse. The digital data architecture should be designed with the ability to ingest such data from various sources – structured/unstructured, batch/ streaming, relational/ non-relational. Not only that, the system should provision for the ability to handle data pipelines for batch, mini-batch or real-time stream processing, enabling instant availability of relevant data for a wide range of business purposes. The agility of data ingestion and processing capability becomes a requirement of extremely high value while exploring and onboarding newly emerging data sources, making the data assets richer and AI/ML applications more effective.

3. **Data storage should be elastic – and expand cost-effectively as data volumes rapidly increase**

Data is one the most important of enterprise assets, but if we don't have a cost-effective infrastructure to store and manage it, it can very quickly become a liability, resulting in either leakage of data due to lack of storage capacity or in heavy expenses and costs to store it.

An effective digital data architecture should not only be flexible, with layered architecture to store raw data, process and analytical data as it grows, but also cost-effective to ensure optimal utilization of resources. The key lies in an effective data storage framework that enables triaging of data assets into hot, warm or cold categories and defining appropriate data storage strategies, leveraging the best of cloud and on-premise storage systems.

4. **Modern master data management**

Master data is the authoritative record of everything vital to an enterprise, including information on users, customers, products, accounts, partners, locations and much more. It is the core discipline of the entire data ecosystem. In the past, MDM required a centralized approach. It worked very well in a stable, not-so-dynamic ecosystem. However, in the digital era, that centralization and rigid schema have made changes and additions difficult and time-consuming, which limits the enterprise's ability to harness value and create dynamic 360-degree views. Modern MDM requires the capability to work across silos, absorb new sources of information, find hidden relationships, quickly generate insights and deliver results in real-time at scale. It should be agile enough to answer any questions that arise, not just those anticipated in advance. At the core of the modern MDM system should lie the ability to understand, identify and manage entity networks and master connect data.

5. **Enable AI/ML and analytics workbench**

The true value of digital data lies in its ability to fuel artificial intelligence and analytics algorithms and help generate insights critical for business decision making and real-time recommendations for engaging customers, mitigating risk and enabling highly efficient operations. The intelligence needs of businesses are very dynamic and, in most cases, cannot be solved by a standard analytics toolset. The digital data architecture should enable curated data for a diverse array of AI, ML, analytics and intelligence toolsets in form of a workbench that can truly democratize the ability to derive and use data insights across different teams in the enterprise, whichever way they want it. An agile AI/ML and analytics workbench also provides the enterprise flexibility to onboard and leverage new,

innovative AI/ML capabilities in the open-source world, hence always bringing the best-of-breed to deliver value.

6. **API-driven, real-time, on-demand data and insight dissemination**

'Real-time' and 'on-demand' are becoming synonymous with the word digital. As the world of data and insights evolved into digital and artificial intelligence, the most fundamental shift was in terms of real-time generation of data and expectation of real-time availability of insights. Today, consumers require instant recommendations, businesses require instant insights, operators require instant alerts . . . and the list goes on. Modern digital data architecture must have as one of its core design principles the question: how do we disseminate insights to the point of action, when users want it and where users want it? A multi-modal real-time insights distribution approach leveraging REST APIs is critical to enable this capability, allowing end-user systems to access insights in real-time and on-demand.

7. **Robust data governance and quality management**

The growth of enterprise data makes its governance extremely important, and it cannot remain a nice-to-have topic anymore. It

is important to have core data governance processes and policies with specific instrumentation for improving data quality across data domains and to have enabling enterprise data transparency through automated data cataloguing. Leveraging AI/ML approaches can help automate the data quality approaches and help uncover hidden data quality issues.

8. **Well defined and rigorously managed data security and regulatory compliance**

As the digital data architecture leverages multi-location (cloud, on-prem) and multi-stack approaches, a well-defined and rigorously implemented data security strategy is necessary. Enterprises need to implement security processes and protocols to manage data security to maintain privacy and security of critical data, network security on cloud, and connectivity to on- prem system, IAM user-group level policies defined with limited access based on work roles and finally AI-enabled proactive monitoring of the data ecosystem.

Nitin's conclusion:

In conclusion, the digital data deluge is unprecedented and very different from the other data challenges that enterprises have handled so far. An innovative and agile approach is needed to realize the opportunities presented by this phenomenon, while also effectively managing the risks it presents. An agile, adaptable and future-ready digital data architecture is a critical capability for enterprises to be able to ride the digital wave. As enterprises embark on or progress in their journey to achieve agile digital data architecture, they need to analyse the value and capabilities of their current data ecosystem and take iterative steps to complement it with the building blocks of digital data architecture.

The digital data architecture is the underlying foundation on which enterprises will need to build their digital journey to achieve transformational business impact and sustainable competitive advantage.

Section 3

Technology Building Blocks of Digital Transformation

Chapter 7

Blockchain - going beyond the hype to Sustainable Value Creation

- As the hype around bitcoin faded away, many read into it the failure of blockchain technology itself. That is untrue. Blockchain is a powerful technology that is just getting off the ground.
- Blockchain offers a new infrastructure-related capability that provides transparency, thereby increasing trust in the system. It has the potential to transform the value chain in many industries. A good example of its usefulness is in payments in banks.
- Blockchain is currently at a nascent stage, with many POCs being tested across industries. It will require a set of challenges to be resolved (outlined in the chapter on the subject) before it becomes a mainstream technology.

Key Takeaways

Blockchain opportunities

The blocks to blockchain implementation

The two-speed strategy for blockchain implementation

Lesson 1: Blockchain: Opportunities and blocks Opportunities with blockchain technology

Blockchain opportunities

- Financial services – Blockchain has wide applications in banks, especially on the payments side. Use cases range from trade finance, derivatives netting and processing of remittances. I see a number of banks now experimenting with blockchain PoCs (proof of concepts). This is similar to the nature of activity we were seeing with AI and machine learning a couple of years back. I expect some of these blockchain use cases in banks to go mainstream in the coming two or three years.
- eGovernance – Digitization is an important objective for many governments, including in India, and blockchain has a very important role to play in this by creating more transparent and accessible public records. For example, blockchain could play a key role in land records and in most areas where certificates need to be issued. In addition to financial services and eGovernance, healthcare and supply chain are two other industries

seeing significant experimentation around blockchain technology's distributed ledger applications.

Blocks to blockchain implementation

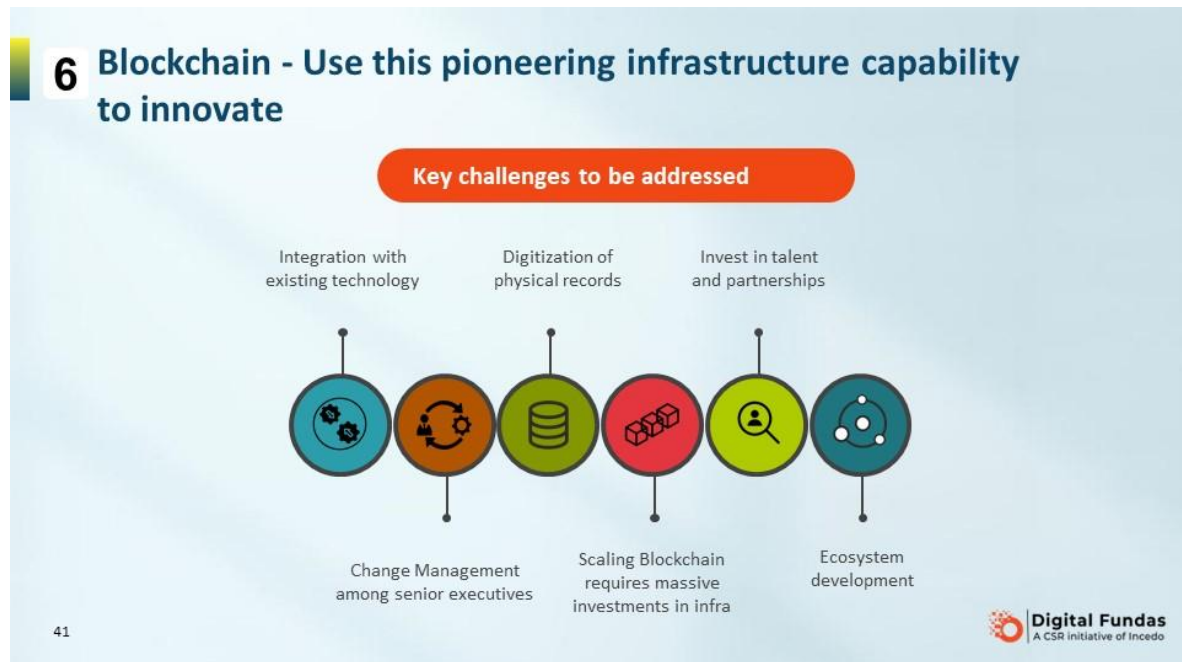
There is a big gap today between the opportunities from blockchain and their realization. A recent survey showed that out of 61 per cent of organizations that are already researching blockchain, only 28 per cent have chosen a blockchain provider. According to the survey, the majority of decision makers are still investigating the technology and comparing vendors, and have not yet defined their stance on blockchain technology.

A TD Bank survey revealed that 90 per cent of treasury and finance professionals think that blockchain and distributed ledger technology (DLT) will positively affect the payments industry. However, in the survey, only 14 per cent of the respondents said their organizations had training strategies for blockchain.

A survey by the Global Blockchain Business Council revealed that 63 per cent of respondents believe that senior business executives have a poor understanding of blockchain technology; 30 per cent consider their knowledge of the emerging technology as 'average'; while the remaining 7 per cent described senior executive understanding of blockchain as 'good'.

The above examples highlight that blockchain is a technology still under development and that there are a number of challenges in its implementation.

The following six key challenges need to be addressed to derive sustainable value from blockchain implementation:



- **Integration with existing technology** – Many of the industries where blockchain technologies are being applied are traditional industries with legacy IT infrastructure. This implies investment in integration efforts that can be time consuming and expensive.

- **Change management** – Understanding of blockchain technologies among senior executives is often low; therefore, sustained evangelization and awareness building is required within organizations.
 - **Digitization of physical records** – Underlying records in many cases are still in physical form (e.g., land records in India) and need to be digitized for blockchain technologies to be applied
 - **Getting blockchain technology to scale** – The blockchain infrastructure is not ready for mass adoption. It requires significant investments, and thus the costs of blockchain adoption are not competitive in some use cases compared with other technology options. This has been particularly true in the payments industry, where despite proven use cases of blockchain, they have not been commercialized at scale because of the costs, especially when compared with the value some of the fintechs have been able to bring.
 - **Talent** – There is a shortage of professionals trained in blockchain technologies, therefore, investment is required in training and in building partnerships,
 - **Ecosystem development** – The vendor and partner ecosystem required to support blockchain technologies and their applications are still developing, therefore, it requires a reality check on use cases that can not only be implemented but also supported, and also careful consideration in choice of partners.
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Nitin's insight

Not too long back a survey reported that advertisers believed 'blockchain' was the most overrated term of 2018. This is hardly a surprise, as many companies from fields completely unrelated to cryptocurrency claimed to have used or tested blockchain technology the year before with little to show for it.

The results of the survey seem to confirm that advertisers were aware of how companies abused the term 'blockchain' just to get free media hype. It was also reported that management consulting firm McKinsey had determined that among the 100-plus supposed use cases for blockchain, the vast majority of pilots and proofs of concept were still stuck in 'pioneering mode' or were being shut down.

Further, blockchain has dropped to the bottom of Gartner Inc.'s annual list of the most disruptive data and analytics capabilities for enterprise IT, having been replaced by tools that leverage artificial intelligence and machine learning.

Is this the end of blockchain? There is certainly a reality check for blockchain, but by no means is it curtains for this technology. However, we need to recognize that blockchain technology is still in a pioneering phase and it will be several years before it becomes dominant.

Lesson 2: The 'two-speed' strategy for blockchain implementation

- **Speed 1: Use case implementation** – Blockchain sometimes feels like a technology searching for problems to solve! It is important to anchor blockchain programs in well-defined use cases where the business pain point and the KPI to be impacted are very clear. That will ensure engagement of business users and provide focus on end-to-end implementation to deliver tangible results.

- **Speed 2: Progressive development of building blocks** – While use cases are the right way to focus and start a blockchain program, they alone are not sufficient. Underlying blockchain technology infrastructure and other key building blocks (e.g., digitization of physical records) have to be developed for sustainable and scalable impact.

Speed 2 of building blocks should run in parallel with speed 1 of use cases, with every sprint of use case implementation further helping to progressively develop the building blocks to the next level of maturity.

Nitin's conclusion:

Blockchain is a nascent technology but one with tremendous potential, given its promise to enhance trust in both record keeping and transactions. Blockchain got hyped well before its time because of the meteoric rise of Bitcoins. Dissipation of the hype around Bitcoins is a good thing, bringing the focus back to the underlying blockchain technology. Many challenges remain, but focused and balanced implementation approaches can help unlock the potential of this powerful technology.

Section 3

Technology Building Blocks of Digital Transformation

Chapter 8

Cloud - an Enabler for Digital Transformation

- Cloud is increasingly becoming the backbone of the technology stack, knitting the whole digital transformation programme together. Earlier, enterprises looked at cloud as a low-hanging fruit for realizing cost take-out in infrastructure, but now the possibilities with cloud have grown significantly.
- Both business units and CIO teams are now looking at cloud for broader purposes than for just making costs variable and infrastructure scalable.
- Business leaders are realizing the value of cloud in reimagining CX and achieving reduced time-to-market as they take on new-age competitors. Whereas CIOs are leveraging cloud for managing enterprise data better and scaling up AI across the enterprise.

Key Takeaways

Cloud as an accelerator for business transformation

Cloud as an enabler for organizational investments

Cloud and Data

Scaling Enterprise AI

Lesson 1: How can the modern cloud accelerate end-to-end digital transformation?

Introduction:

As part of their digital transformation journey, organizations are reimagining customer experiences, innovation, productivity and data strategy (and this list is incomplete). In fact, the digital natives have demonstrated that cloud is integral to execution along all these dimensions, and can become a powerful catalyst for not just revenue growth but also for driving competitive advantage through digital transformation.

The track record of cloud usage outside of the digital natives has been less than stellar: It is estimated that 27 per cent of organizations' cloud computing budget was deemed to have gone to waste in 2019. What can we learn from how the digital natives leveraged the cloud to drive competitive advantage?

We start by looking at two specific areas where the cloud can improve business performance.

Cloud as accelerator for business transformation

1. **Reimagined customer experiences**

Digital experience is clearly table stakes across industries. Consumers expect a complete digital self-service experience with access to their information, the facility to conduct transactions anytime, anywhere, from any device. In response, firms are having to rethink their entire customer workflows from ground up in an effort to get to an ideal digital state – one that ranges across email, mobile, website, push notifications, voice and more. In doing so, there is an opportunity to leverage cloud as a key enabler:

- **Bridging the legacy technology barrier:** Putting together a seamless digital user experience on the legacy technology backbone is not just daunting, in many cases it is plainly impossible. The cloud has provided a way for firms to put together a digital experience without having to completely upgrade their ageing infrastructure. This is especially true for firms with age- old installations of core systems. Building a digital stack on that infrastructure is clearly a non-starter. The cloud provides a way to initiate a digital user experience even as the legacy technology modernization is playing the catch-up game.
- **Integration of next-generation technologies:** Delivering a seamless digital experience to customers requires integration of multiple next-generation technologies, from digital assistants (e.g., chatbots) to image recognition (e.g., digital verification). Building these technologies and integrating them into the customer workflows within the existing technology infrastructure using the in-house talent is often impossible. On the other hand, all the major cloud providers offer pre-packaged solutions, many of them with AI-cognitive technologies trained over the years to a high degree of accuracy. For instance, UPS leveraged Microsoft's cloud-based Azure bot service to deploy a chatbot in its mobile app, which enables customers to engage in text-based and voice-based conversations to get the information they need about shipments, rates, and UPS locations.

2. **Reduced time to market**

Traditional firms would do well to take a leaf out of the e-commerce playbook and look to the cloud to reduce their time to market – both when it comes to expanding their products and services to include digital channels as well as in designing and launching digital-first services. Here are some tips from their playbook:

- **Partnering for non-core activities** and focusing on the core product development process reduces the time and cost to market. The traditional methods of discovery of product features, price points, etc., with focus groups have always been slow, expensive and often end up being hit-or-miss. Firms are increasingly transitioning this process to digital experiments as a proxy for understanding the product market fit. A leading telecom service provider adopted such a strategy to discover the market-price fit for wearable accessories as a medical assistant device for senior citizens. They ran a series of A/B experiments by partnering with a cloud-based provider, which helped them design and execute the experiments by integrating with the telecom provider's e-commerce portal. A process that would normally take several months was brought down to a few weeks before the service was launched at the national level.
- **Build vs. buy vs. subscribe** is an important choice that determines an enterprise's ability to onboard and offer services rapidly and seamlessly and, where possible, subscription-based solutions offered by best-in-class providers can dramatically cut the time to market. Medical providers have been on the tele-health journey for several years now. Delivering quality care through a digital channel requires integration of multiple technologies: high-fidelity video interactions in a secure manner and remote device management to measure and track medical data, to name a few. These services by themselves require sophisticated technology, not to mention putting them together to create a seamless experience for both doctor and patient. Needless to say, most medical providers have opted for cloud-based solutions to take these services to market.
- **Responding to market signals with speed:** The need and urgency for reliable digital solutions have never been as compelling as in the post-Covid world, which saw a sharp

increase in digital engagement in a matter of a few weeks. A recent statistic: while the average share of customer interactions that are digital grew from 20 per cent (June 2017) to 36 per cent (December 2019), this jumped from 36 per cent (December 2019) to 58 per cent (July 2020), accelerated by the Covid-19 pandemic. And this is not a temporary spike either – this is likely to be the ‘new normal’. Companies across industries have struggled to scale their infrastructure to respond to this rapid switch to digital channels – except the ones that were already on a cloud infrastructure. If nothing else, this has led to an acceleration of cloud adoption.

- Most firms have just begun to scratch the surface of the scale, speed and innovation potential that cloud has to offer in driving digital transformation.

Cloud as an enabler for organizational investments

1. Data

Data quality: Several technology vendors in data integration (e.g., Talend, Informatica) offer the ability to deploy rule-based and AI/ ML-based data profiling and quality methodologies that must be applied to data at the point of ingestion from the source systems. This is especially useful in cataloguing newer data sources, which tend to be unstructured (e.g., call transcripts), and high volumes of noisy data (e.g., clickstream data).

Data integration: Interfaces from legacy systems to the cloud must be developed with a clear objective to integrate data from different sources based on well-established ontologies. This is easier said than done – especially when there is an accumulation of decades of historical data. This is why most firms are opting to define data integration on a forward-looking basis, without getting bogged down by historical data integration issues.

Cloud data governance: Migration to the cloud brings the opportunity to redefine cloud governance – including compliance and regulatory oversight. There are a slew of firms that offer technology-based governance solutions such as data cataloguing (e.g., Alation) and metadata management (e.g., Collibra) to drive data governance at the time of data ingestion.

2. Scaling enterprise AI

AI model training and validation: With data on the cloud, it becomes much more efficient and effective to train and operationalize models in production environments. The availability of cloud compute and storage is essential for training/validating the models with ever-expanding data sources.

AI model features store: Creating a centralized cloud-native features store helps data science teams across the enterprise to reuse existing features. Over time, this will promote reuse.

AI model training at scale: AI algorithms are usually data and compute intensive. The elastic nature of cloud can be very useful in scaling up/down of compute capacity to handle the training and validation of models in a variable cost structure.

AI best practices: Most cloud AI platforms include open source algorithms that have been curated over the years. Access to a cloud AI platform enables access to these best practices.

AI integration with operational systems: Most cloud AI platforms support API-based integration with most of the leading enterprise transaction systems (e.g., ERPs) and decision support systems (e.g., workflow management systems). This helps reduce the time to operationalize model recommendations for driving decision making.

Nitin's conclusion

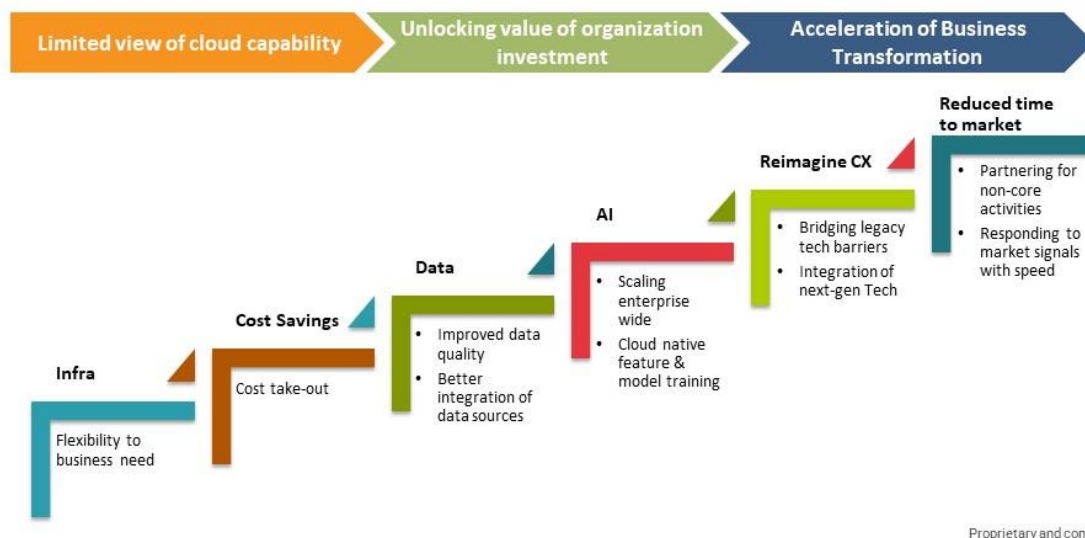
Over the last twenty-five years, IT organizations have gone through several technology hype cycles – many of them were just that. If there were any lingering doubts about cloud as just another technology fad, the pivotal role of cloud in enabling the hyper-acceleration of digital enablement across companies triggered by the Covid-19 pandemic have laid these doubts to

rest. It is now clear that cloud represents business-driven value, and with that the focus is shifting to how to capture that value. The focus now needs to be on ensuring that companies are not moving too slowly towards the cloud on their digital transformation journey in the post-Covid-19 'next normal' phase, running the risk of disruption from digital-first, nimbler competition.

Lesson 2: Cloud, a key enabler of digital transformation: 6 Mantras

Cloud - Core enabler for Digital Transformation

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- **Treat cloud as more than an IT initiative – it is a joint effort with business:** For the power of cloud to be fully brought to bear, joint ownership between business and IT will help identify and prioritize programmes that directly impact business outcomes.
- **Redesign the governance processes:** In a cloud-first world, firms will have to redefine the governance processes to handle the data security, regulatory and compliance requirements without losing out on the flexibility and agility of a cloud infrastructure. This requires a complete redesign of the legacy governing processes to create a cloud-first data, security and infrastructure governance organization.
Embrace the extended cloud technology ecosystem: A cloud- based infrastructure opens up access to best-in-class applications at every level in the technology stack through the cloud marketplaces. Organizations must take a relook at their 'build vs. buy' strategies for applications.
- **Do not get bogged down by the 'myth of a single source of truth':** Enterprises continue to pour millions into chasing the 'golden record' as part of cloud migration efforts, which often ends up as an empty promise. It is often better to focus on forward-looking data quality and harmonization at the point of data ingestion, and let the quality of data improve over time.
- **It is all about the people:** Over the years, firms have built large IT teams around legacy technologies. The talent and organizational model within IT and data organizations needs to be evaluated on an ongoing basis.
- **From project to platform and program mindset:** Once the leadership team recognizes the strategic importance of cloud, it is important to orient all stakeholders towards programs and platforms rather than projects. Projects are typically one-time investments for point problems and are driven by short-term priorities. Cloud platforms, in particular,

require stable, ongoing focus and sponsorship with consistent ownership over a multi-year period.

Nitin's conclusion

Over the last twenty-five years, IT organizations have gone through several technology hype cycles – many of them were just that. If there were any lingering doubts about cloud as just another technology fad, the pivotal role of cloud in enabling the hyper-acceleration of digital enablement across companies triggered by the Covid-19 pandemic have laid these doubts to rest. It is now clear that cloud represents business-driven value, and with that the focus is shifting to how to capture that value. The focus now needs to be on ensuring that companies are not moving too slowly towards the cloud on their digital transformation journey in the post-Covid-19 'next normal' phase, running the risk of disruption from digital-first, nimbler competition.